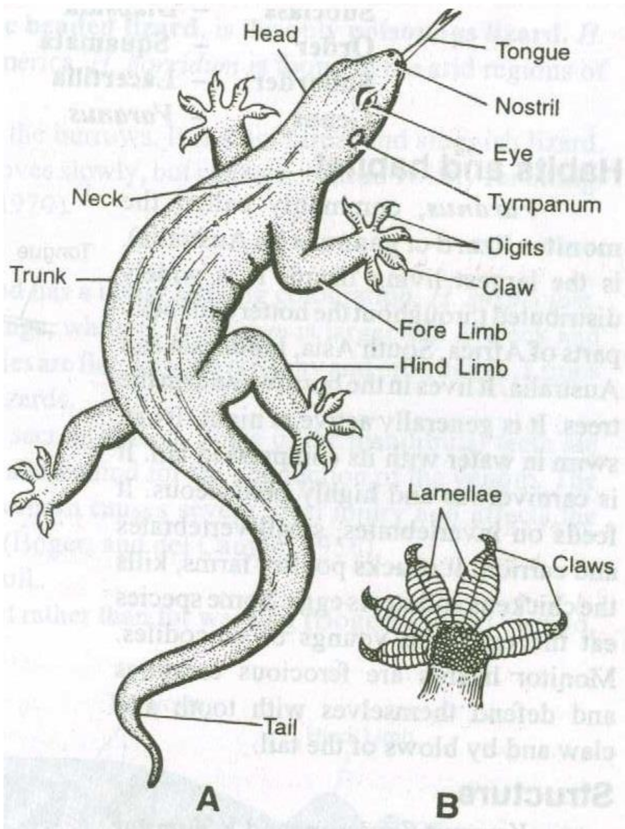
CLASS REPTILIA

There is no simple definition for this group of animals because there is no single characteristic that is unique to reptiles. Generally, they are vertebrates that have the following features:

- ✓ Reptiles are amniotes that have evolved a cleidoic egg.
- ✓ Shape of body may be compact in some and elongated in others.
- ✓ Body covered with an exoskeleton of dry, horny, cornified skin or iniegument (not slimy) usually with epidermal scales or scutes on the surface. Bony dermal plates present in some. Skin has few glands.
- ✓ Limbs paired usually with five digits or toes, each ending in a horny claw.Limbs suited for crawling, running or climbing or in marine turtles for swimming. They are rreduced in some lizards and are absent in a few other lizards and in all snakes.
- ✓ Endoskeleton well ossified. Skull monocondylar (only one occipital condyle is present to articulate with the first vertebrae).
- ✓ Teeth homodont (all of the same type).
- ✓ Ribs with sternum (sternum absent in snakes) forming a complete thoracic basket. Two sacral vertebrae support the pelvic girdle.
- ✓ In most reptiles the heart is either three chambered or imperfectly four chambered. The auricle is divided into two chambers the right and the left and the ventricle is only partially divided. Crocodiles, as a result of complete separation of ventricle have four chambered heart. In reptiles, there is separation of oxygenated (pure/ arterial) and non-oxygenated (impure/venous) blood in the heart.
- ✓ Cutaneous respiration is negligible. Reptiles respire mainly by lungs which are ventilated by changing body size. Gills are absent in reptiles. Pharyngeal and cloacal respiration is used in some aquatic turtles. In embryonic life, respiration is by branchial arches.
- ✓ Reptiles are ectothermic. However, they maintain a relatively high body temperature during their periods of activity by behaviourally controlling their exposure to sun. They are more active than amphibians
- ✓ Excretion by means of paired metanephric kidneys; uric acid is the main nitrogenous waste excreted.
- ✓ Nervous system is provided with optic lobes on the dorsal side of the brain; and 12 pairs of cranial nerves.
- ✓ Paired tympanic ears (*tympanic membrane, is the technical term for the eardrum. Anatomically, it functions as a thin, tightly stretched diaphragm that vibrates when struck by airborne sound waves*.) have independently evolved in most reptiles from amphibians. They have been secondarily lost in some.

- ✓ Sexes are separate, fertilization internal usually by copulatory organ found in males.
- ✓ Eggs are large with much yolk and are covered by leathery or calcareous shells; usually laid and incubated outside but retained within the body for development by some lizards and snakes. Extra - embryonic membranes (amnion, chorion, yolk sac and allantois) present during development. Young ones on hatching (or on being born) resemble adults. Metamorphosis absent in reptiles.
- ✓

✓ 

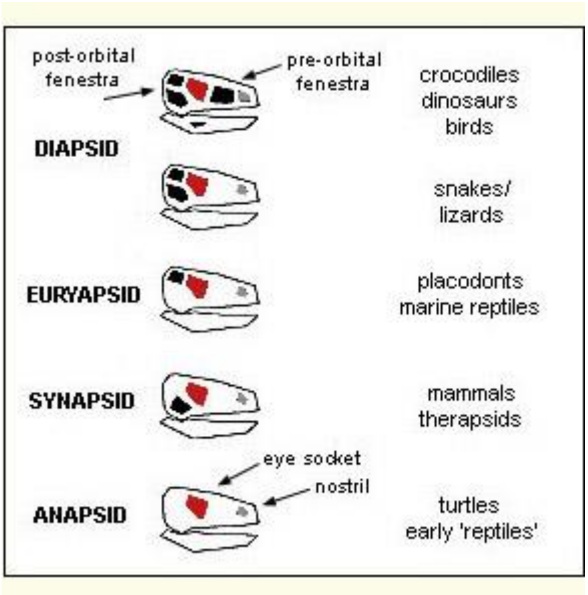
A – Dorsal View B – Ventral Side of the foot

Hemidactylus (House or Wall Lizard)

Evolution and Adaptive Radiation of Reptiles

Since they were the first vertebrates to be fully terrestrial, they had few competitors and they multiplied rapidly and occupied all available niches and specialized accordingly. Their adaptive radiation involved different methods of locomotion and feeding. Different feeding methods resulted in modification of jaw muscles and the temporal regions of the skull and so reptiles are grouped according to the morphology of the skull.

- In the most primitive reptiles, the roof of the skull has openings for the major sense organs only; nostril, eyes and pineal eye. Such a condition is known as anapsid and reptiles with that kind of skull are placed in subclass Anapsida. Example of anapsid reptiles are the turtles.
- In other reptiles there are other openings in the temporal region known as temporal fenestrae. Those with a single opening high on each side of the cheek are put in subclass Euryapsida.
- Those with a single opening on the side but more ventrally placed than in euryapsids belong to subclass Synapsida, the mammal like reptiles that are all extinct. They gave rise to the mammals.
- In diapsid condition, both pairs of fenestrae occur. There two subclasses with this type of condition, Lepidosauria (for example snakes) and Archosauria (example crocodiles). They give rise to birds and other reptiles like snakes and crocodiles.



Temporal openings of different evolutionary lines of Vertebrates

Order Testudines

Chelonia: Turtles

- ✓ These are the only surviving anapsids.
- ✓ Body in a bony case of dermal plates, consisting of a dorsal carapace and a ventral plastron;
- ✓ jaws with horny beaks instead of teeth; vertebrae and ribs fused to overlying carapace;
- ✓ tongue not extensible;
- ✓ neck usually retractable;
- ✓ Tortoises are completely terrestrial and have dome-shaped shells that predators find difficult to bite, crush or swallow.

Diapsids

All other reptiles are diapsids. They belong to three orders: Squamata, Crocodilia and Sphenodonta.

Order Sphenodonta/Rhynchocephalia): tautaras

- ✓ These were abundant in the early Mesozoic era but today only two species survive; *Sphenodon punctatus* and *S. guntheri.*
- ✓ Until recently it was thought that only the former species existed. They are confined in distribution to New Zealand.
- ✓ They look like lizards but have a more primitive skull structure than any true lizard.
- ✓ They move using legs, lateral undulation of the trunk and tail.
- ✓ They have a pineal eye on top of the head. Males have no distinct copulatory organ and transfer of sperms is by pressing the cloacas together.
- ✓ They are referred to as the "living fossils" because they have not changed much form the species that exist 150 million years ago.

Order Squamata;

Lizards

- ✓ Lizards and snakes belong to order Squamata; lizards are the oldest and most primitive squamates. They are placed in suborder Lacertilia (or Sauria).
- ✓ Many species are slender, some compressed laterally and flattened dorso-ventrally.
- ✓ Limbs may be long or short, are reduced on some and entirely lacking in. a number of limbless lizards.
- ✓ All limbless lizards live in the soil. The tail vertebrae in many species are incompletely ossified.
- ✓ If the tail is seized the vertebrae separate, the animal escapes and in time regenerates another tail.

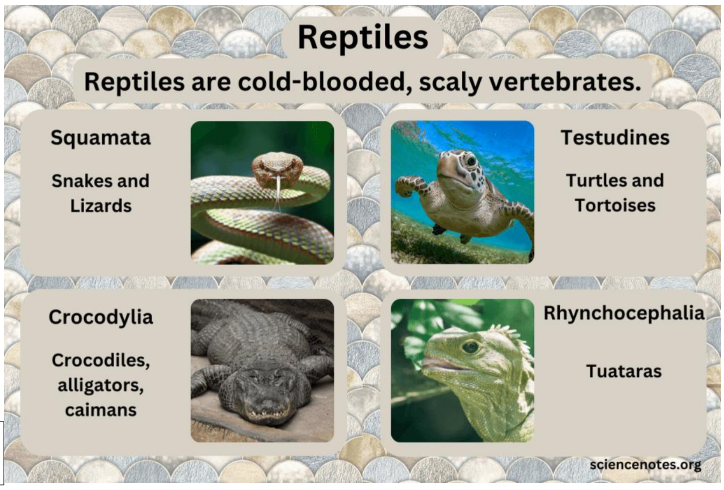
Snakes

- ✓ They belong in order Squamata, suborder Ophidia.
- ✓ They have elongate body.
- ✓ They have no limbs or girdles.
- ✓ The sternum, eyelids, external ear openings and bladder are also lacking.
- ✓ The eyes are covered by a nictitating membrane.

- ✓ Their jaws are extensively flexible and can swallow prey much bigger than a mammal of the same size.
- ✓ They can detect low frequency vibrations through their skull bones. They compensate for lack of senses and sight and hearing with a very sensitive tongue.
- ✓ The skull is delicate and most of the bones can move upon one another. Teeth slant backwards and are fixed on jaws and on bones on the roof of the mouth. Thy are used to hold food while being swallowed Venomous snakes have a pair of specialized teeth or fangs on the lower maxillary bone.
- ✓ Snakes are feared because of their venom or poison. Venom is saliva containing a mixture of digestive enzymes such as proteases, collagenases and phospholipases. Venom kills and starts the digestion.
- ✓ In some snakes the fangs are at the front part of the upper jaw and they inject venom into the prey before it is swallowed.
- ✓ In others, the fangs are towards the back of the mouth and venom is injected as the prey is swallowed. Some of the venoms work by affecting the nervous system (neuro toxic venom) and kill very fast by paralyzing the prey. Others have poison that affects the blood and tissues (heamotoxic venom), causing heamolysis, clotting and so on. Many snakes (about $^3/_4$), however, are not poisonous but suffer because most of us in Africa believe that a snake once spotted should not be left alive. There are two species of lizards that are poisonous.

Order Crocodilia: Crocodiles and alligators

- ✓ These are the most advanced reptiles.
- ✓ The heart is completely divided into four chambers.
- ✓ Though they are poikilotherms, their body temperature is closer to constant than in other reptiles.

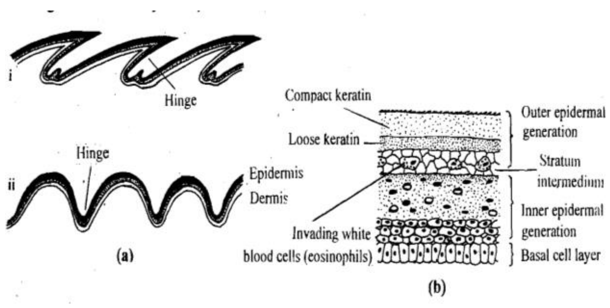


Adaptations of Reptiles to Terrestrial Life

Reptiles were the first vertebrates to truly break away from aquatic dependencies. They achieved complete terrestrial independence through a suite of critical physiological, structural, and behavioral adaptations.

❖ Integument (body covering)

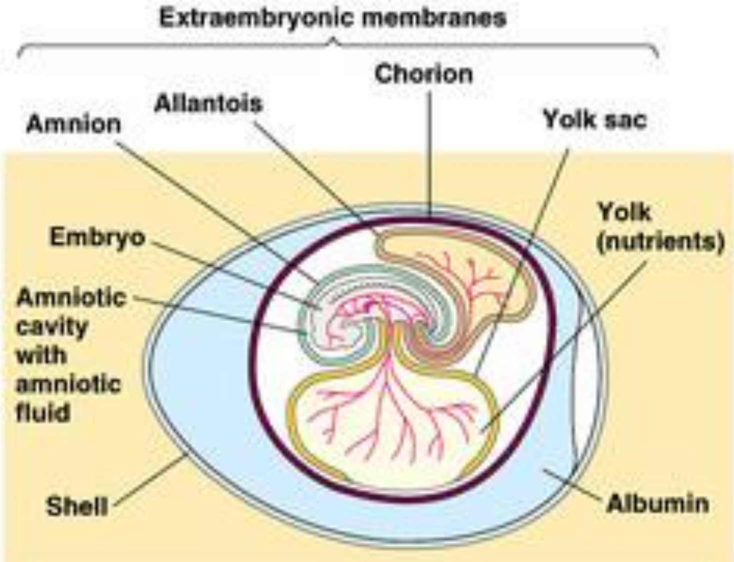
- ✓ The epidermis is dry and covered with horny scales with large amount of keratin, a water-insoluble protein deposited by epidermal cells.
- ✓ One difference between the scales of reptiles and those of fishes and amphibians is that they arise from the dermis rather than from the epidermis.
- ✓ The scaly skin protects the animal from desiccation, mechanical damage and harmful radiation.
- ✓ There are few glands on the skin although some have scent glands that produce secretions used in species and sexual recognition.
- ✓ In some snakes and lizards glands secretes irritating or sticky material that protects them from predators. Bone may develop in the dermis beneath the bony scales, as is the case with the bony plates in a turtle shell and small nodules of bone in some lizards and crocodiles.



❖ The Amniotic Egg

- ✓ One of the most important achievements of reptiles was the evolution of the cleidoic egg, a self-- contained system (Gr kleidoun = to lock up).
- ✓ The egg is provided with all the necessary food material and with fats that when metabolized yield water for the developing embryo's needs.
- ✓ The eggs obtain additional moisture through the shell and so most reptiles deposit their eggs in moist places.
- ✓ This egg is enclosed within a shell that provides protection and reduces the chances of desiccation and yet permits exchange of gases.

- ✓ Most reptilian eggs have leathery shells that are more permeable to carbon dioxide and oxygen than the hard shells of birds' eggs.
- ✓ Embryonic waste materials are stored in the eggs. Well-formed cleidoic eggs are found in reptiles, birds, insects and egg-laying mammals.



the cleidoic egg

In viviparous reptiles and mammals that give birth to live young, the eggs have little or no shell but the egg extra-embryonic membranes, are still present. Four extra-embryonic membranes are present. They lie outside of the embryo and play an important role in protecting and maintaining the embryo. They are;

- Yolk sac that surrounds the yolk mass.
- Amnion encloses the embryo and is filled with fluid which protects the developing embryo. Reptiles, birds and mammals are called amniotes because they all have this membrane.
- Chorion lies beneath the shell and surrounds both the yolk sac and amnion.
- Allantois grows out of the posterior end of the gut and comes to lie against the inner side of the chorion). Blood from the embryo is circulated through the wall of the allantois by allantoic vessels (called umbilical vessels in mammals). Gaseous exchange occurs here. Nitrogenous wastes in form of uric acid crystals are deposited in the cavity of allantois.

The yolk sac, allantois and amnion all connect to the centre of the belly region of the embryo and the connecting cord is homologous to the umbilical cord of mammals. The yolk sac is reabsorbed during the course of development and the connection with other extra embryonic membranes is broken when the newborn hatches from the egg. The membranes are left behind with the shell.

The cleidoic egg enables the reptiles to live in drier habitats than amphibians and many are found in deserts Some have readapted to aquatic life but all lay their eggs on land unless they can broad them internally, like in sea snake. They are abundant in the tropics, a few are in temperate regions and are absent in arctic regions.

❖ Skeleto-muscular System, and Locomotion

Reptiles have well ossified endoskeletons which are stronger than that of the amphibians. The skull is monocondylar.

Teeth are homodont and fused to jaw bones. An reptiles except the limbless members have better body support than amphibians and more efficiently designed limbs for travel on land.

Many of the dinosaurs walked on powerful hind limbs alone. The thrust of hind legs is transferred to the vertebral column through two sacral vertebrae instead of a single one as found in amphibians.

Feet have toes with claws which allow them to get a good grip upon the substratum.

❖ Jaws

The reptilian jaws are more efficiently designed for using crushing force on the prey.

Jaws of fish and amphibians are designed for fast jaw closure, but once the prey is caught, little static force can be used.

In reptiles, however jaw muscles are longer, larger , and arranged for more efficient mechanical advantage.

❖ Behavioral Thermoregulation

Reptiles as you know are ectothermic, gaining body heat from their surroundings.

They are not endothermic like birds and mammals which generate their own heat by metabolic processes.

Body temperature of reptiles living in temperate regions falls sharply at night, making them sluggish. On sunrise, they seek out sunny spots and warm up their bodies by solar radiation and by conduction of heat from sunwarmed rocks and soils on which they bask.

Absorption of solar energy is further enhanced early in the morning by the dispersal of pigment in the chormatophores of skin so that the skin darkens and absorbs more light energy and also by an increased blood flow through the skin. When it becomes too warm the skin lightens, cutaneous blood vessels get constricted and the animals move towards shade. A few species such as chuckwalla (Sauromalus) can also cool their body by eliminating heat through panting.

Reptiles(*heliotherms*) thus can maintain a high and relatively constant body temperature and remain active during daytime by behavioral thermoregulation, (adjusting their exposure to the sun).

Some of the lizards in United States maintain, even when air temperature is considerably lower, temperatures of about 34°C (93°F) which is close to that of most mammals.

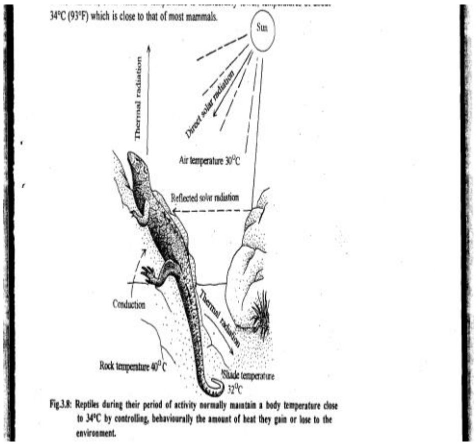


Fig.3.8: Reptiles during their period of activity normally maintain a body temperature close to 34°C by controlling, behaviourally the amount of heat they gain or lose to the environment.

❖ Feeding and Digestion

Low food requirements allow reptiles to live in deserts and other habitats where there is not enough food to sustain high 'energy demands of many mammals.

For example a frog or a mouse would be enough to provide a small snake the energy it needs for a week or more. Most reptiles except for tortoise and some lizards are carnivourous.Some have specialized ways of catching and swallowing food.

❖ Respiratory Adaptations

Well-developed lungs: Reptiles are born with lungs, unlike amphibians that rely partly on gills. This allows efficient oxygen uptake in dry environments.

Expanded rib cage: Facilitates ventilation and supports larger lung capacity.

❖ Excretion

- ✓ Reptiles like other amniotes have metanephric kidneys.
- ✓ Less water is needed to remove nitrogenous waste from the blood than is amphibians because they produce mainly uric acid in forming crystals that do not require water to remove.
- ✓ Some of the water removed by the kidney is reabsorbed by other parts of the kidney tubules, the urinary bladder and cloaca.